

The Solar Cycle – Data Locations and Guidance

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This document provides guidance on downloading and plotting the data for The Solar Cycle class.

Google Trends:

<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>

- Type 'aurora australis' into the search bar
- Change the location to 'Worldwide'
- Change the time to '2004 - present' (Google began collecting trends data in 2004)
- Leave the search type as 'Web Search'
- To download the data as a .csv file, press the 'Download CSV' icon at the top right of the plot
- The data is presented as relative search interest per month

Note: The **Trove archive** data cannot be freely accessed. An access key (available from Trove upon request) is required. Copyright restrictions apply to most content younger than 70 years old.

Canberra Geomagnetic Observatory Data: https://imag-data.bgs.ac.uk/GIN_V1/GINForms2

- Select CNB (Canberra Australia) as the Observatory
- Data can be downloaded in 1-year (366 day) chunks from 01-01-1991 (start of INTERMAGNET data availability) to present
- Leave data type as 'Best available' and sample period as 'Minute data'
- IAGA-2002 is the recommended file type
- Data will need to be processed in a Jupyter notebook (or similar) to isolate the geomagnetic x component

aa index: https://isgi.unistra.fr/data_download.php

- Select the data type as *aa* (since 1868)
- You will need to download the data in two chunks (there is a 100-year download limit):
 - 1868-01-01 to 1968-01-01
 - 1968-01-01 to present date
- Set the output data format as 'IAGA2002-like'
- Enter your email address in order to download the data
- Data will be split into definitive (D), provisional (P) and quick-look (Q) files depending on its age and whether it has been reviewed
- Data downloads as a ZIP file which you will need to extract the files from
- Files will download as .DAT files that you can view with a text editor
- You will need to combine all the data files as a single file in a Jupyter notebook (or similar) before it can be plotted
- The raw plotted data is noisy and trends are unclear
- To view trends clearly, monthly means and yearly (smoothed) means should be plotted

Sunspot Number: <https://www.sidc.be/SILSO/datafiles>

- Download the 'Yearly mean total sunspot number [1700 - now]' as a .csv file
- Data is presented as yearly averages

Solar Cycle Boundaries: <https://sidc.be/SILSO/cyclesminmax>

- Solar cycle boundaries can be downloaded as a .txt or .csv file from the SILSO website

Plotting advice:

The data for this class can be plotted up in Microsoft Excel or in a Jupyter notebook (using pandas and matplotlib) depending on the experience of the students.

Use yearly averages where possible and consider smoothing trends (such as with the aa data) to make trends clearer.

References:

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<https://geomagnetism.ga.gov.au/geomagnetic-indices/aa-index>

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